AIRLINE PERFORMANCE AND REVENUE ANALYSIS

USING POWER BI

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Data Visualization

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1. Introduction

This project focuses on analyzing airline performance, passenger trends, revenue generation, customer ratings, and flight delays using Power BI dashboards. The dashboard helps visualize airline operations and provides business insights for better decision-making.

2. Objectives

- Analyze airline revenue trends
- Study passenger growth
- Evaluate customer ratings

- Analyze flight delays
- Compare domestic and international routes
- Build interactive dashboards

3. Tools and Technologies Used

Tool	Purpose
Power BI	Dashboard Development
Excel	Data Source
DAX	Calculations
Power Query	Data Cleaning

4. Dataset Description

The dataset contains airline operational and financial data including flights, revenue, passengers, delays, ratings, and route types.

Column Name	**Description**
Flight_ID	Unique flight ID
Revenue_USD	Revenue generated
Passengers	Passenger count
Delay_Minutes	Flight delay

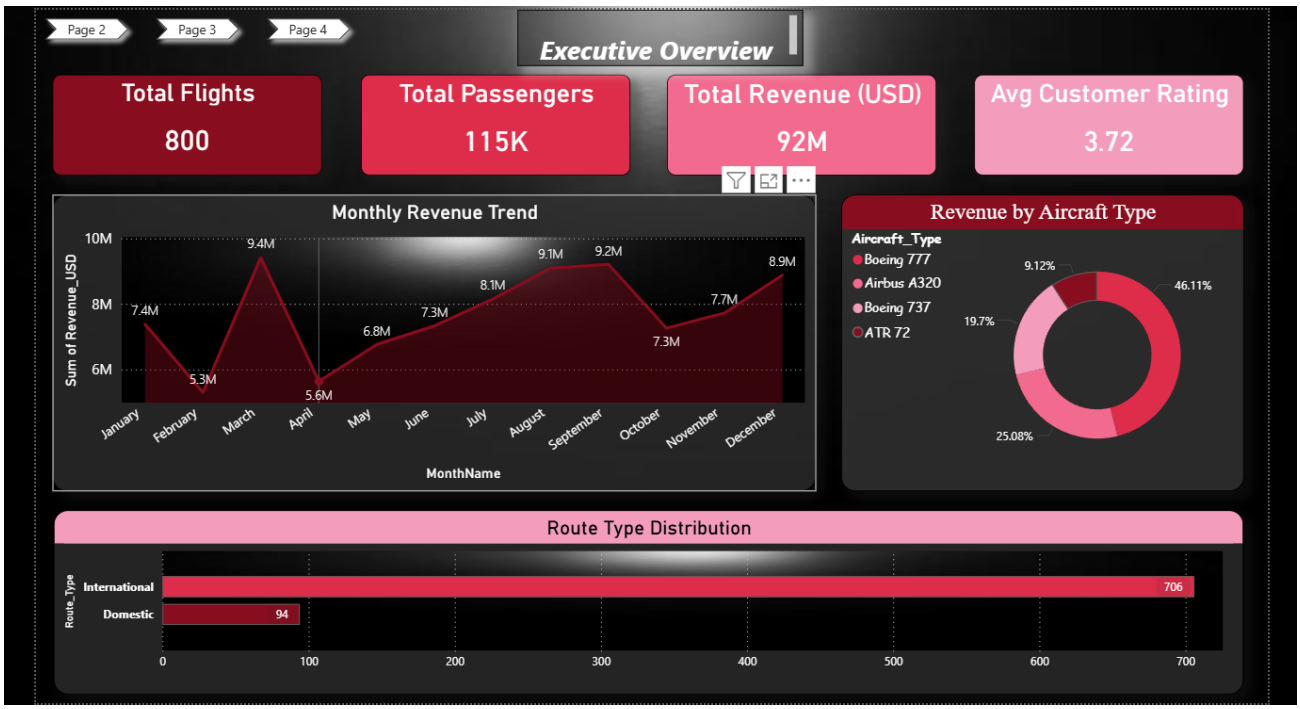
Column Name	Description
Customer_Rating	Passenger feedback
Aircraft_Type	Aircraft model
Route_Type	Domestic/International

5. Data Cleaning and Transformation

- Removed null values
- Corrected data types
- Created calculated columns
- Applied DAXX measures
- Used Power Query transformations
- Created MonthName olumn

6. Dashboard Pages

6.1Executive Overview



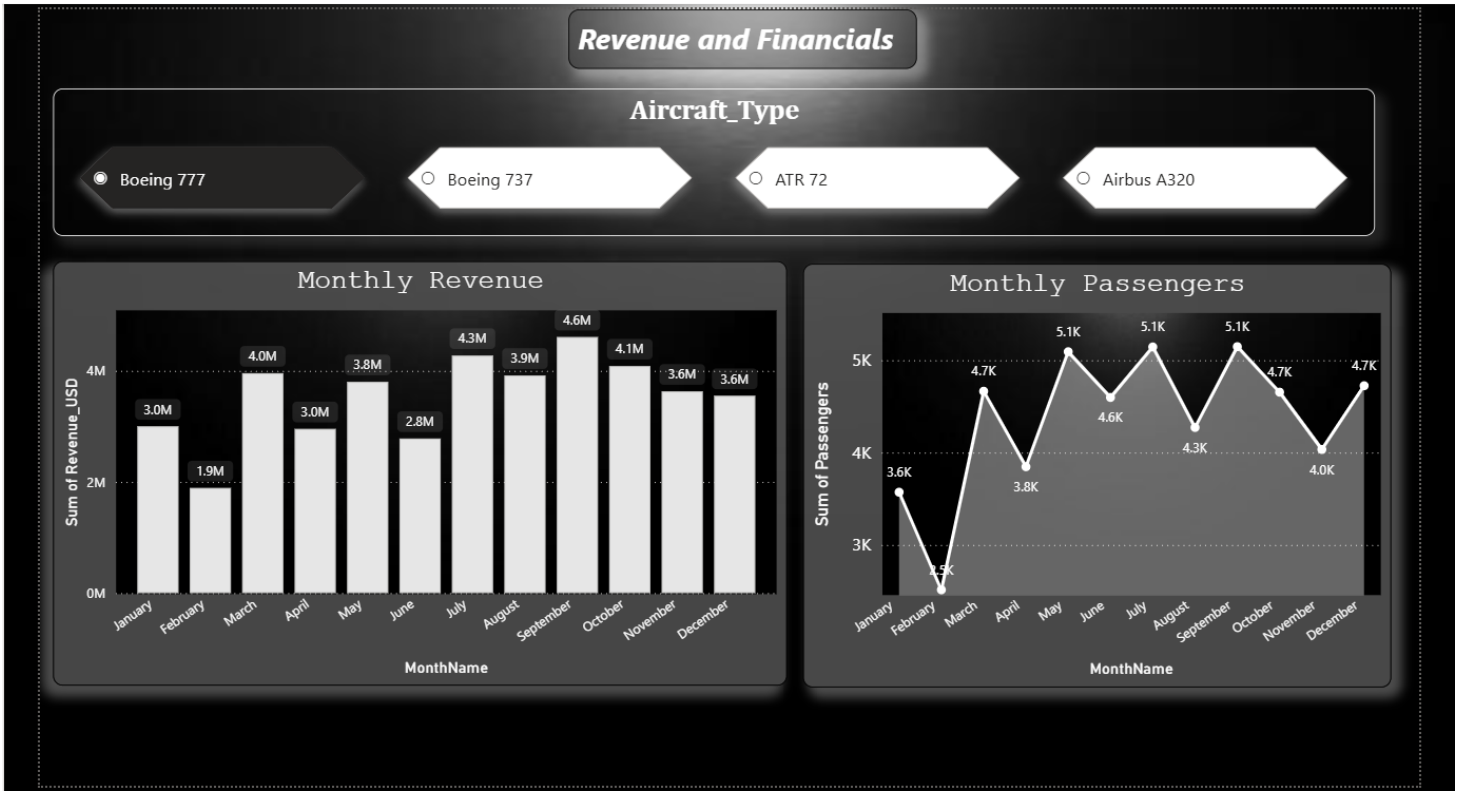
KPIs

- Total Flights: 800
- Total Passengers: 115K
- Total Revenue: 92M
- Avg Customer Rating: 3.72

Insights

- Revenue peaked in March
- International routes dominate
- Boeing 777 contributes highest revenue

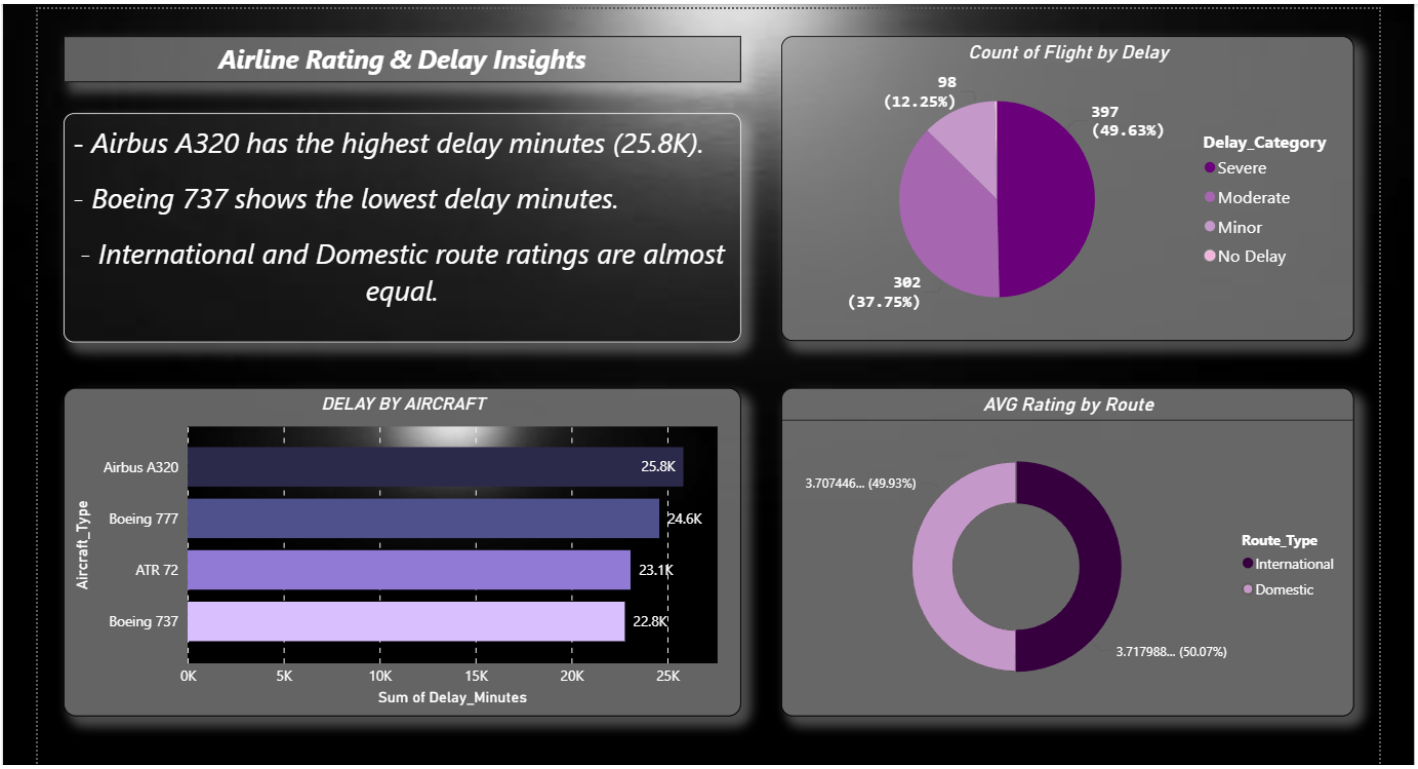
6.2 Revenue and Financials



- Monthly revenue analysis

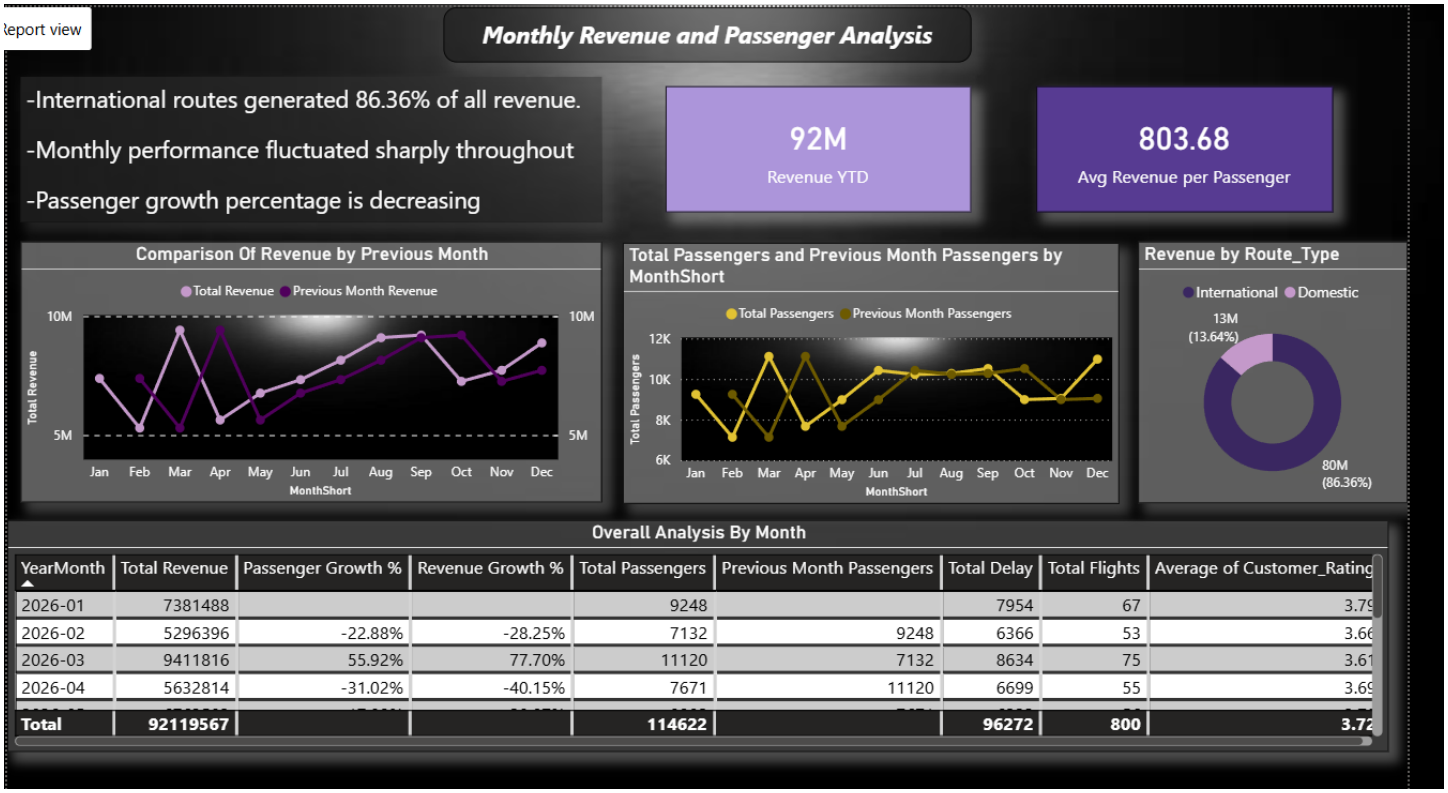
- Aircraft-wise performance
- Passenger trends

6.3 Airline Rating and Delay Insights



- Airbus A320 has highest delays
- Boeing 737 has lowest delays
- Delay categories analyzed

6.4 Monthly Revenue and Passenger Analysis



YearMonth	Total Revenue	Passenger Growth %	Revenue Growth %	Total Passengers	Previous Month Passengers	Total Delay	Total Flights	Average of Customer_Rating
2026-01	7381488			9248		7954	67	3.79
2026-02	5296396	-22.88%	-28.25%	7132	9248	6366	53	3.66
2026-03	9411816	55.92%	77.70%	11120	7132	8634	75	3.61
2026-04	5632814	-31.02%	-40.15%	7671	11120	6699	55	3.69
Total	**92119567**			**114622**		**96272**	**800**	**3.72**

- International routes generate most revenue
- Monthly growth comparison
- Revenue fluctuation analysis

7. DAX Measures

Total Revenue = SUM(Airline[Revenue_USD])

Total Passengers = SUM(Airline[Passengers])

Avg Rating = AVERAGE(Airline[Customer_Rating])

8. Challenges Faced

- Handling large data
- Building accurate DAX measures
- Dashboard formatting
- Maintaining consistent theme

9. Conclusion

The project successfully analyzes airline operations using Power BI dashboards. Interactive visualizations help identify trends in revenue, passengers, delays, and customer ratings, enabling data-driven decision-making.

10. Future Scope

- Real-time data integration
- Predictive Analytics
- AI-based forecasting
- Mobile dashboard optimization

11. References

https://powerbi.microsoft.com

Dataset link:

https://www.kaggle.com/datasets/guriya79/pia-airlines